

Edgar Gabriel Caballero Torres

Data Scientist | Project Manager

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Summary

Data Scientist with experience in predictive and time series modeling, ETL pipeline development in Google Cloud, and design of scalable interactive dashboards. Specialized in Python (Pandas, NumPy, Scikit-learn, Statsmodels, and TensorFlow), SQL, and visualization tools (Power BI), applying advanced Machine Learning techniques, data analytics, and statistics to generate actionable insights and optimize decision-making. I am recognized for clear and structured communication, analytical thinking, complex problem-solving, innovation, self-management, and proactivity. Effective collaborator in hybrid and changing environments, with constant curiosity and a focus on continuous improvement.

Skills

Programming & Query Languages: Python, SQL, JSON

Data Science & Machine Learning: Machine Learning, Scikit-learn, TensorFlow

Data Analysis & Visualization: Pandas | Polars, NumPy, Matplotlib | Seaborn, Jupyter Notebook, Power BI

Data Engineering & Infrastructure: ETL (Google Cloud Dataflow), APIs

Development & Tools: Git/GitHub

Languages: English (Professional), Spanish (Native)

Soft Skills

Clear and structured Communication (oral and written) | Critical Thinking | Problem Solving | Innovation | Creativity | Self-Management | Proactivity | Collaboration (in-person and remote) | Curiosity and Continuous Learning | Adaptability | Constructive Conflict Management | Entrepreneurial mindset | Communicate and collaborate with cross-functional teams

Experience

Paredes y Asociados Constructores S.A. de C.V., Construction Site Supervisor | Mexico City, Mexico
Project Designer | Site Manager June 2023 – June 2025

- Leadership | Led the supervision of a \$35 million MXN (USD 1.75M) project, ensuring compliance with specifications.
- Analysis | Implemented Revit-AutoCAD and Revit-BIMx workflows to review and execute-supervise client-driven changes.
- Management | Coordinated multidisciplinary teams for timely project delivery, demonstrating excellent problem-solving capability.

Ibero Steel, Construction Site Supervisor | Project Designer Mexico City, Mexico
Dec 2022 – June 2023

- Planning and Structural Design | Led the comprehensive design of 4+ standard padel courts, optimizing material selection and manufacturing workflows.
- Visualization | Generated 10+ marketing renders using V-Ray and SketchUp, increasing client approval rates.
- Technical Documentation | Developed assembly manuals for prefabricated components, reducing on-site and manufacturing errors.

Projects

Gold Recovery Prediction | Industrial Process Optimization

Regression models to predict gold recovery rates in a mining flotation process.

- Engineered features on 16K industrial measurements to predict recovery; optimized Random Forest/Gradient Boosting models using custom sMAPE.
- Identified key drivers (particle size, reagent concentration) to maximize gold yield, providing actionable process recommendations.
- **Skills used:** Regression Modeling, Feature Engineering, Ensemble Methods, Python (Pandas, Scikit-learn).

Taxi Demand Forecasting | Time Series Analysis

Time series forecasting models to predict hourly airport taxi demand.

- Engineered temporal features (lags, rolling statistics) and performed time series decomposition to identify key daily/weekly patterns and peak demand hours.
- Developed and optimized a hybrid forecasting pipeline (ARIMA & ensemble models), achieving <10% RMSE for 1-hour-ahead predictions to inform driver allocation.
- **Skills used:** Time Series Analysis, ARIMA Modeling, Feature Engineering, Seasonal Decomposition, Demand Forecasting, Temporal Pattern Recognition.

A-B Test Evaluation | Conversion Impact

Designed and analyzed an A/B experiment to measure the impact of a new variant on user click-through conversions.

- Defined control (con) and experimental (exp) groups at the user level and validated random assignment and sample balance before analysis.
- Computed and compared click-through rates between groups, quantifying the absolute and relative uplift in conversions achieved by the experimental variant.
- **Skills used:** A/B Testing, Experimental Design, Hypothesis Testing, Conversion Rate Analysis, Statistical Inference, Data Cleaning & Aggregation.

Education

Cert. Tripleten, Data Science

Intensive, project-based program covering the full data science lifecycle, from data wrangling to deploying machine learning models for business impact

- Completed 20+ hands-on projects and 600+ hours of curriculum, mastering Python, SQL, ML, and cloud tools to solve real-world business problems.

Mexico City, Mexico
Mar 2025 – Nov 2025

BS Universidad Veracruzana, Architecture

Bachelor's degree in Architecture.

Mexico City, Mexico
Sept 2014 – Nov 2019

Certifications

Data Science Methodology | IBM

2023

Mastered the CRISP-DM framework.

- Applied the end-to-end CRISP-DM methodology: from business understanding and data preparation to modeling, evaluation, and deployment planning.

Python for Data Science, AI & Development | IBM

2024

Python foundation for data science and AI.

- Mastered Python programming (functions, OOP, error handling) and core data science libraries (Pandas, NumPy) for data manipulation and analysis.

Construction Project Management | Columbia University

2022

The fundamentals of construction management, LEAN projects and sustainability

- Mastered management for construction industry, the role of project manager, contract types, and project delivery methods.